

ZigBee Technology Architecture and Its Applications

In this present communication world, there are numerous high data rate communication standards that are available, but none of these meet the sensors' and control devices' communication standards. These high-data-rate communication standards require low-latency and low-energy consumption even at lower bandwidths. The available proprietary wireless systems' Zigbee technology is low-cost and low-power consumption and its excellent and superb characteristics make this communication best suited for several embedded applications, industrial control, and home automation, and so on. The Zigbee technology range for transmission distances mainly ranges from 10 – 100 meters based on the output of power as well as environmental characteristics.

What is Zigbee Technology?

Zigbee communication is specially built for control and sensor networks on IEEE 802.15.4 standard for wireless personal area networks (WPANs), and it is the product from Zigbee alliance. This communication standard defines physical and Media Access Control (MAC) layers to handle many devices at low-data rates. These Zigbee's WPANs operate at 868 MHz, 902–928MHz, and 2.4 GHz frequencies. The data rate of 250 kbps is best suited for periodic as well as intermediate two-way transmission of data between sensors and controllers.

Zigbee is a low-cost and low-powered mesh network widely deployed for controlling and monitoring applications where it covers 10–100 meters within the range. This communication system is less expensive and simpler than the other proprietary short-range wireless sensor networks as Bluetooth and Wi-Fi.

Zigbee Architecture

Zigbee system structure consists of three different types of devices as Zigbee Coordinator, Router, and End device. Every Zigbee network must consist of at least one coordinator which acts as a root and bridge of the network. The coordinator is responsible for handling and storing the information while performing receiving and transmitting data operations.

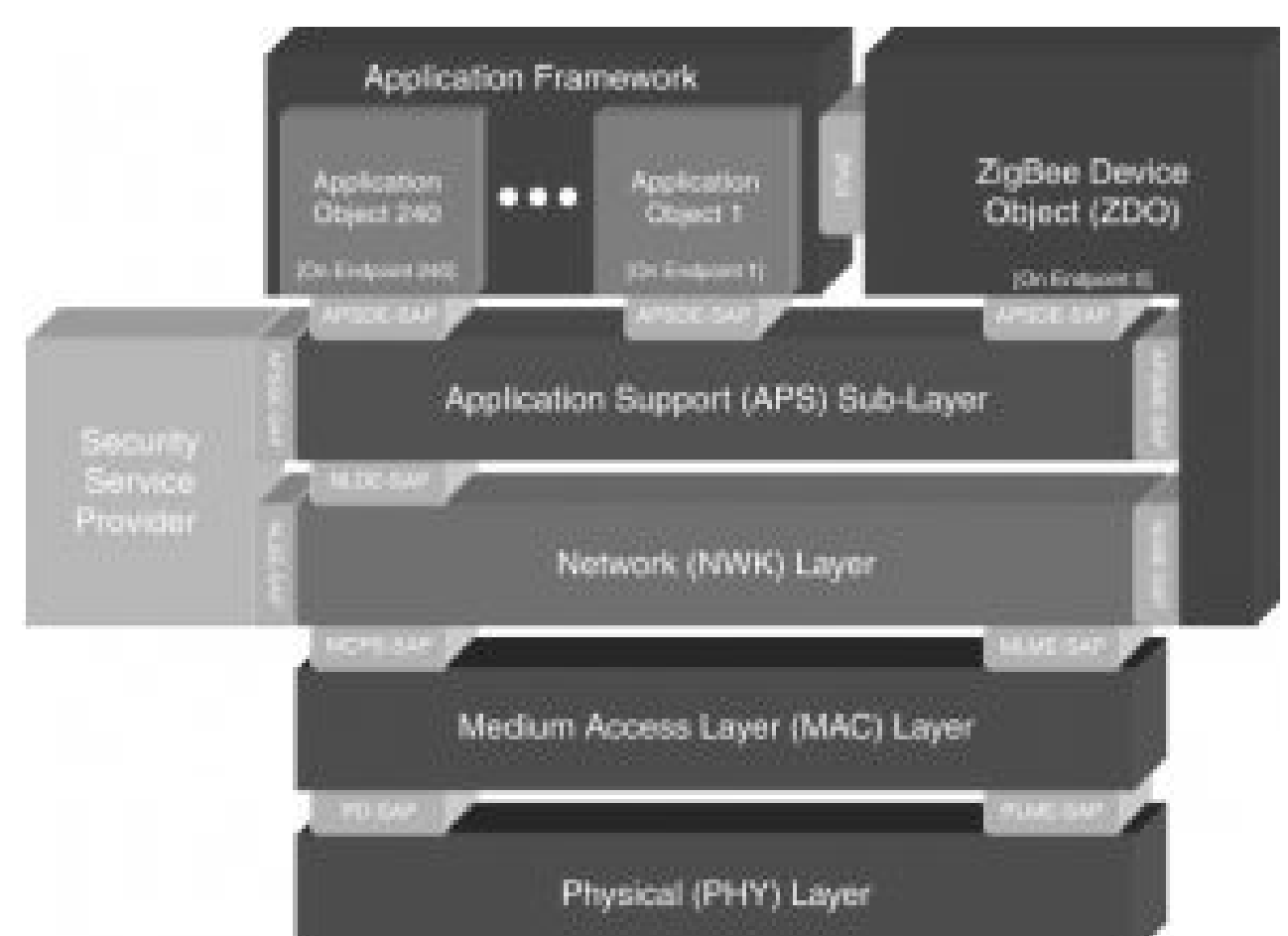
Zigbee routers act as intermediary devices that permit data to pass back and forth through them to other devices. End devices have limited functionality to communicate with the parent nodes such that the battery power is saved as shown in the figure. The number of routers, coordinators, and end devices depends on the type of networks such as star, tree, and mesh networks.

Zigbee protocol architecture consists of a stack of various layers where IEEE 802.15.4 is defined by physical and MAC layers while this protocol is completed by accumulating Zigbee's own network and application layers.

ZigBee Technology Architecture

Physical Layer: This layer does modulation and demodulation operations upon transmitting and receiving signals respectively. This layer's frequency, data rate, and a number of channels are given below.

MAC Layer: This layer is responsible for reliable transmission of data by accessing different networks with the carrier sense multiple access collision avoidances (CSMA). This also transmits the beacon frames for synchronizing communication.



Network Layer: This layer takes care of all network-related operations such as network setup, end device connection, and disconnection to network, routing, device configurations, etc.

Application Support Sub-Layer: This layer enables the services necessary for Zigbee device objects and application objects to interface with the network layers for data managing services. This layer is responsible for matching two devices according to their services and needs.

Application Framework: It provides two types of data services as key-value pair and generic message services. The generic message is a developer-defined structure, whereas the key-value pair is used for getting attributes within the application objects. ZDO provides an interface between application objects and the APS layer in Zigbee devices. It is responsible for detecting, initiating, and binding other devices to the network.

Zigbee Operating Modes and Its Topologies

Zigbee two-way data is transferred in two modes: Non-beacon mode and Beacon mode. In a beacon mode, the coordinators and routers continuously monitor the active state of incoming data hence more power is consumed. In this mode, the routers and coordinators do not sleep because at any time any node can wake up and communicate.

However, it requires more power supply and its overall power consumption is low because most of the devices are in an inactive state for over long periods in the network. In a beacon mode, when there is no data communication from end devices, then the routers and coordinators enter into a sleep state. Periodically this coordinator wakes up and transmits the beacons to the routers in the network.

These beacon networks are work for time slots which means, they operate when the communication needed results in lower duty cycles and longer battery usage. These beacon and non-beacon modes of Zigbee can manage periodic (sensors data), intermittent (Light switches), and repetitive data types.

Zigbee Topologies

Zigbee supports several network topologies; however, the most commonly used configurations are star, mesh, and cluster tree topologies. Any topology consists of one or more coordinators. In a star topology, the network consists of one coordinator which is responsible for initiating and managing the devices over the network. All other devices are called end devices that directly communicate with the coordinator.

This is used in industries where all the endpoint devices are needed to communicate with the central controller, and this topology is simple and easy to deploy. In mesh and tree topologies, the Zigbee network is extended with several routers where the coordinator is responsible for starting them. These structures allow any device to communicate with any other adjacent node for providing redundancy to the data.

If any node fails, the information is routed automatically to other devices by these topologies. As redundancy is the main factor in industries, hence mesh topology is mostly used. In a cluster-tree network, each cluster consists of a coordinator with leaf nodes, and these coordinators are connected to the parent coordinator which initiates the entire network.

Due to the advantages of Zigbee technology like low cost and low power operating modes and its topologies, this short-range communication technology is best suited for several applications compared to other proprietary communications, such as Bluetooth, Wi-Fi, etc. some of these comparisons such as range of Zigbee, standards, etc., are given below.

The advantages of Zigbee include the following.

- This network has a flexible network structure
- Battery life is good.

- Power consumption is less
- Very simple to fix.
- It supports approximately 6500 nodes.
- Less cost.
- It is self-healing as well as more reliable.
- Network setting is very easy as well as simple.
- Loads are evenly distributed across the network because it doesn't include a central controller
- Home appliances monitoring as well as controlling is extremely simple using remote
- The network is scalable and it is easy to add/remove ZigBee end device to the network.

The disadvantages of Zigbee include the following.

- It needs the system information to control Zigbee based devices for the owner.
- As compared with WiFi, it is not secure.
- The high replacement cost once any issue happens within Zigbee based home appliances
- The transmission rate of the Zigbee is less
- It does not include several end devices.
- It is so highly risky to be used for official private information.
- It is not used as an outdoor wireless communication system because it has less coverage limit.
- Similar to other types of wireless systems, this ZigBee communication system is prone to be bothered from unauthorized people.

Applications of Zigbee Technology

The applications of ZigBee technology include the following.

Industrial Automation: In manufacturing and production industries, a communication link continually monitors various parameters and critical equipment. Hence Zigbee considerably reduces this communication cost as well as optimizes the control process for greater reliability.

Home Automation: Zigbee is perfectly suited for controlling home appliances remotely as a lighting system control, appliance control, heating, and cooling system control, safety equipment operations and control, surveillance, and so on.

Smart Metering: Zigbee remote operations in smart metering include energy consumption response, pricing support, security over power theft, etc.

Smart Grid monitoring: Zigbee operations in this smart grid involve remote temperature monitoring, fault locating, reactive power management, and so on.

ZigBee technology is used to build engineering projects like wireless fingerprint attendance system and home automation.